Experiment 32: Charge Oscillations and Energy Transfers in LRC Circuits

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This study's primary objective is to first, calculate critical values for resistance that lead to three defining configurations for a damped, discharging LRC circuit: normal, critical, and oscillatory damped cases; and second, demonstrate the relationship between electric and magnetic field energy exchange including losses from heat dissipation. Critical resistance values of the normal, critical, oscillatory cases were 1133 Ω, 485 Ω, and 20 Ω, respectively. Analysis on data collected by LabPro sensors and software demonstrated close agreement with the oscillatory case, with an experimental damped frequency of 46 Hz and a theoretical damped frequency of 47.1 ± 1.6 Hz. Energy analysis confirmed the continuous exchange between electric and magnetic field energy, however, erroneous conserved energy plots were produced from sensitive signal triggering in the software calibration. More appropriate trigger values and higher sample frequencies would help mitigate this problem and produce more accurate plots of conserved energy.

I. INTRODUCTION

Inductor, resistor, capacitor (LRC) circuits are a fundamental configuration of components that allows us to examine the behaviour between electricity and magnetism. This is because capacitors store energy in an electric field, and inductors store energy in its generated magnetic field from the change in current, and a discharged circuit demonstrates the energy transfer between them. Understanding LRC circuits is fundamental to modern electronics, and is responsible for radio frequency communications from integrated antennae in PCBs, motor drivers, and capacitive touch sensors, for example.

From Kirchoff's voltage law, the primary differential equation that models the discharge behaviour is

$$L\frac{\mathrm{d}^2 i}{\mathrm{d}t^2} + R\frac{\mathrm{d}i}{\mathrm{d}t} + \frac{i}{C} = 0. \tag{1}$$

Where L, R, C, i are the inductance, system resistance, capacitance, and current, respectively. The general solutions to this equation can be reduced to three configurations that display differing behaviour by selecting values for R: Normal , critical, and oscillatory damping; Eqs. (2), (3), and (4), respectively.

$$\left(\frac{R}{2L}\right)^2 - \frac{1}{LC} > 0 \tag{2}$$

$$\left(\frac{R}{2L}\right)^2 - \frac{1}{LC} = 0 \tag{3}$$

$$\left(\frac{R}{2L}\right)^2 - \frac{1}{LC} < 0 \tag{4}$$

For the critically damped case, theoretical predictions can be made on the voltage signals. The theoretical equations for current and voltage are given by the following:

$$i = \frac{V_0}{L} t e^{-\frac{R}{2L}t}. \tag{5}$$

$$V_L = -V_0\left(1 - \frac{R}{2L}t\right)e^{-\frac{R}{2L}t}. \tag{6}$$

$$V_C = V_0\left(1 + \frac{2}{RC}t\right)e^{-\frac{R}{2L}t}. \tag{7}$$

V_0, V_L, V_C are the values for the steady state voltage, inductor, capacitor electric potentials, respectively. For the oscillatory case, the theoretical frequency of oscillation is given by

$$j\omega = \sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}}. \tag{8}$$

The energetics of the system can be calculated using potential energy equations:

$$E_L = \frac{1}{2}Li^2 \tag{9}$$

$$E_C = \frac{1}{2}CV_C^2 \tag{10}$$

And the thermal heat energy dissipated is calculated using

$$E_H = \int_{t_n}^{t_{n+\Delta t}} Ri^2 \,\mathrm{d}t \approx Ri_{AV}^2 \Delta t, \tag{11}$$

where Δt is the evaluated time step between samples, and i_{AV} is the average current over the time interval given by

$$i_{AV} = \frac{V_R}{R_{res}}. \tag{12}$$

II. APPARATUS

Equipment: Vernier LabPro sensor interface, LabPro software, 4000 turn inductor, electrolytic capacitor, decade resistance box, differential voltage probes, Instek GFG-8219A function generator, Wavetek LCR55 digital multimeter with alligator clip probes, banana plug cables.

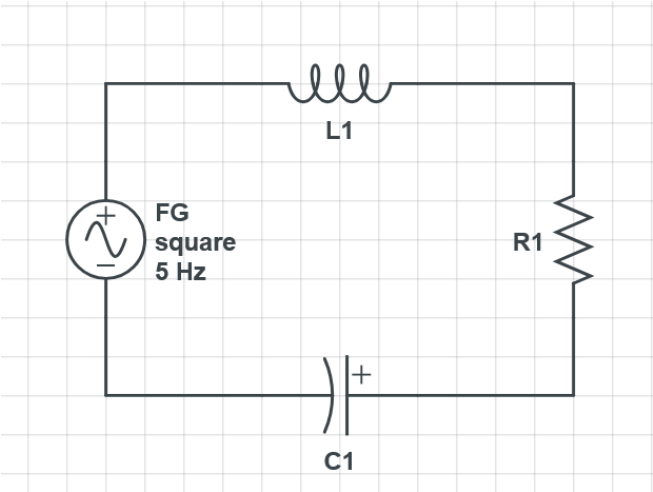


FIG. 1. LRC circuit. Probes were connected across L, R, C components to measure voltage in the LabPro Software.

III. PROCEDURE

The experimental apparatus and procedure followed the standard method outlined in the University of Victoria PHYS 326 laboratory manual [1].

A circuit comprised of an inductor, resistor, and capacitor (LRC circuit) was assembled in series connected to a function generator. Measurements of the inductance, resistance, and capacitance were taken of each component with the digital multimeter before connecting the circuit. Table I provides a list of measurements taken. Uncertainty was calculated using the digital multimeter's spec sheet, including the range's relative and per digit uncertainty.

TABLE I. Experimental Parameters

Quantity	Reading
R_L, Inductor resistance	$109.4 \pm 1.3\,\Omega$
L, Inductance	$1.044 \pm 0.072\,\mathrm{H}$
C, Capacitance	$10.07 \pm 0.21\,\mu\mathrm{F}$
R_{FG}, function generator resistance	$50.00 \pm 0.25\,\Omega$
R_{res}, normal	$1133\,\Omega$
R_{res}, critical	$485\,\Omega$
R_{res}, oscillatory	$20\,\Omega$

The function generator was configured to output a 5 Hz square wave signal to act as a continuous switch for the LRC circuit to quickly cycle between charge and discharge. The function generator offset was manually adjusted such that the "ON" and "OFF" settings were set to 3.86 V and 0 V, respectively. Vernier differential probes were connected to the LabPro sensor interface and the LabPro software was calibrated to take voltage readings at 3000 samples per second.

Before each data were collected, care was taken to ensure the probes were both physically shorted and zeroed in the software to mitigate charge buildup that might offset the readings. A preliminary test case was done to adjust the maximum recording time so that the majority of the discharge cycle was captured without extending to the charging cycle. The maximum time was set to 0.09 seconds, which appeared to consistently capture the whole event. Resistor values were calculated to define damped, critically damped, and oscillatory behaviour for the LRC circuit. The critical resistance value for the damped case was set to be double that of the critically damped case to ensure signal behaviour between the two cases were appropriately different. Once the decade resistance box was set using the dials, a discharge cycle was recorded via a voltage trigger and values for inductor, resistor, and capacitor voltages produced. This was done a total of three times, completed once for each damped, critically damped, and oscillatory case.

IV. RESULTS

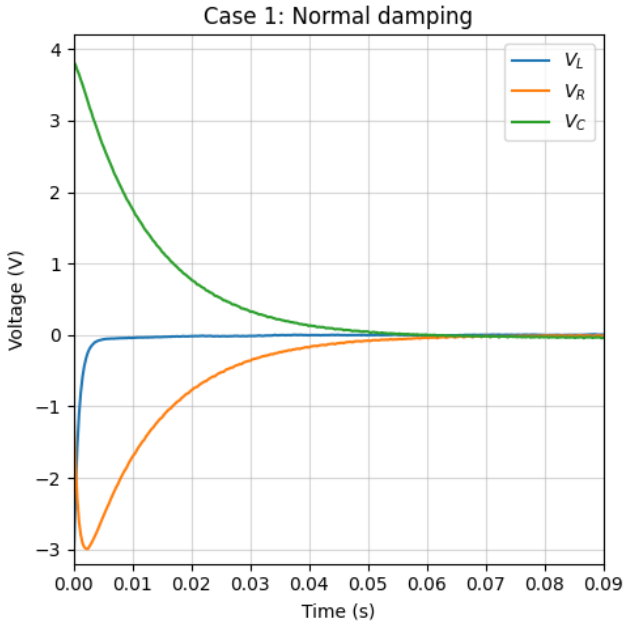


FIG. 2. Signals showing normal damping behaviour.

In Figure 2, the circuit is normally damped such that the components discharge gradually over the cycle to zero

volts without oscillating. The resistance value chosen here was calculated using double the critical system resistance found in Eq. (3), which ended up being 1133 Ω.

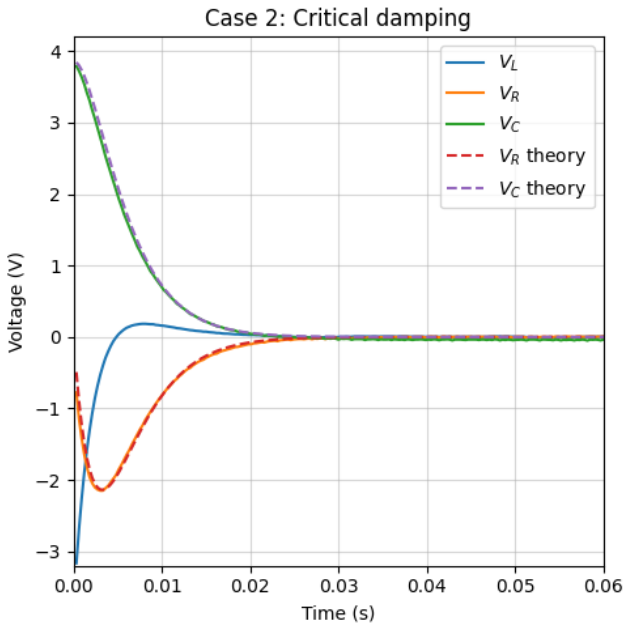


FIG. 3. Signals showing critical damping behaviour.

In Figure 3, the resistance chosen using Eq. (3) was 485 Ω. Notice the quicker time to reach zero volts compared with case 1.

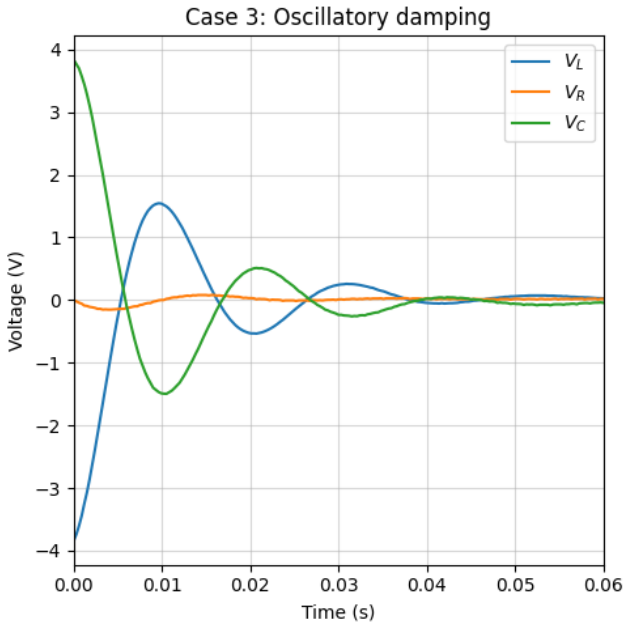


FIG. 4. Signals showing oscillatory damping.

Figure 4 shows that at selected resistance 20 Ω, the behaviour turned oscillatory. Since the resistance was much lower, more of the electric potential remained within the inductor and capacitor during the whole discharge cycle.

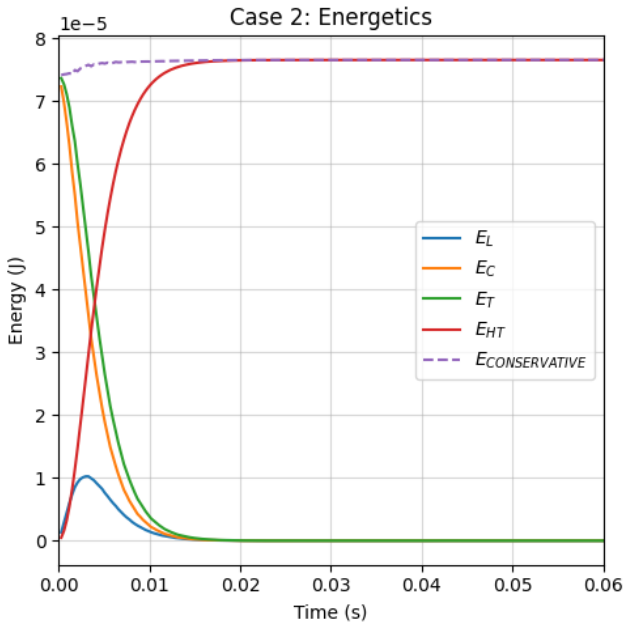


FIG. 5. Critically damped energetics

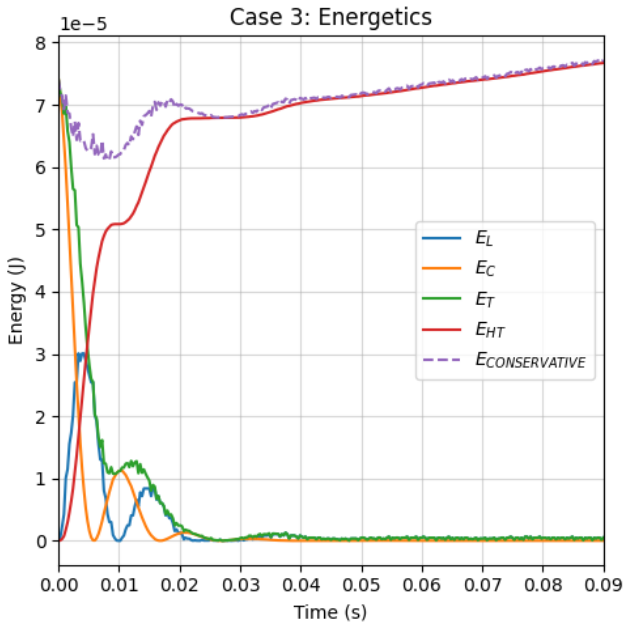


FIG. 6. Oscillatory energetics

Figures 5 and 6 demonstrate the energetics of the system in critically damped and oscillatory cases. In both figures, it shows that once the discharge is initiated, the stored energy transfers between the electric and magnetic fields of the circuit. As E_T decreases, the accumulated

dissipated thermal energy, E_{HT}, rises due to joule heating.

TABLE II. Oscillation Frequency

Quantity	Value	Units
Experimental frequency	46	Hz
Theoretical natural frequency	49.1 ± 1.8	Hz
Theoretical frequency incl. R	47.1 ± 1.6	Hz

The oscillation frequency results are listed in Table II. Experimental frequency was determined by measuring the peak-peak time and dividing by the period. Theoretical values were calculated using Eq. (8). The theoretical natural frequency is close, but not quite consistent with the determined value, however, the theoretical frequency taking into account R is consistent. This suggests the small damping resistance produces a system frequency lower than the natural frequency. Note: Uncertainties in values were calculated using the aid of the *uncertainties* python package.

V. DISCUSSION

Figures 2, 3, and 4 show the voltage signals of the components agree with the theoretical predictions. The critically damped system discharges quicker than the normally damped configuration, and the oscillatory system shows the electric potential transferring between the magnetic and electric fields. Furthermore, the theoretical models for V_R and V_C align almost exactly to the recorded experimental signals. Small differences between the experimental and theoretical results here can be explained by the resolution of the evaluated time step, Δt. As Δt converges to zero, the difference between the theory and experiment would decrease. The V_R and V_C signals attenuate much faster in the critically damped case. This is due to the current decreasing faster as a result of the reduced resistance in the circuit.

However, in examining the energetics in Figures 5 and 6, the conserved energy unexpectedly rose above the initial total energy. This suggests the sum of cumulative heat dissipation energy and total energy did not agree with theoretical predictions. Since no external energy sources existed in the circuit, the increase in energy can be explained by a few contributing factors. In the oscillatory case, the energy transfer between electric and magnetic fields decreased in magnitude over time but never reached zero during the cycle. In addition, the accumulated heat dissipation, E_{HT}, increased rapidly and unexpectedly rose higher than the initial energy of the system. This can be explained by the fact that the whole cycle was not captured from initial discharge, but indeed began slightly after due to the software's trigger settings.

E_T and E_{HT} show to have correlated signals in both cases. When E_T drops, E_{HT} rises, and when E_T begins to plateau, so does E_{HT}. This is because the total energy of the components transferred into the current through the system, thus resulting in joule heating. In Case 2, E_{HT} plateaus quickly because current rapidly attenuates from critically damped discharge. In Case 3, E_{HT} gradually plateaus because the induced current from oscillations takes longer to attenuate.

Also, though a small contribution, the model does not take into account the thermal heat dissipated through the wires connecting the circuit, only that of the system impedance of the individual components. As temperatures change, so do the internal resistances of the components, further affecting the calculated result due to changing current flow.

Analysing the frequency response of the oscillatory case, we found that the experimentally determined frequency was consistent with the theoretical frequency taking into account the system resistance, R. The theoretical natural frequency of the system was found to be slightly higher and just outside consistency margins. This suggests the damped frequency is slightly lower than the natural frequency of the system due to the contributing damping resistance, R.

VI. CONCLUSION

From our experimental analysis, the transient response of the LRC circuit under normal, critical, and oscillatory damped configurations demonstrated the relationship between electric and magnetic fields during a discharge event. The electric potential clearly showed that energy was exchanged back and forth between the components, and the experimental signals strongly agreed with the theoretical prediction curves. In addition, the frequency calculations were consistent with experimental observations.

The energetics plots demonstrated an unexpected increase in conserved energy, but this was explained by the data being captured too far ahead of the initial discharge event. This erroneous reduction in the initial total energy stored in the circuit led to a pseudo upward trend in the conserved energy. By adjusting the trigger settings to be more sensitive to the falling edge of the signal, a full capture of the discharge cycle, from fully charged to fully discharged, would have shown a higher initial total energy, and thus shown the conservative energy to gradually decrease.

Appendix A: Sample Calculations

Critical resistance values were calculated via

$$\begin{aligned} R_{calc} &= 2 \cdot \sqrt{\left(\frac{L}{C}\right)} \\ &= 2 \cdot \sqrt{\left(\frac{(1.044 \pm 0.072) \times 10^0}{(1.007 \pm 0.021) \times 10^{-5}}\right)} \\ &= 644 \pm 23\,\Omega \end{aligned}$$

$$\begin{aligned} R_{res} &= R_{calc} - R_L - R_{fg} \\ &= (6.440 \pm 0.233) \times 10^2 - (1.094 \pm 0.013) \times 10^2 - (5.000 \pm 0.025) \times 10^1 \\ &= 484 \pm 23\,\Omega \end{aligned}$$

The theoretical natural frequency is

$$\begin{aligned} f_{theo} &= \left(\frac{1}{2 \cdot \pi}\right) \cdot \sqrt{\left(\frac{1}{L \cdot C}\right)} \\ &= \left(\frac{1}{2 \cdot \pi}\right) \cdot \sqrt{\left(\frac{1}{(1.044 \pm 0.072) \cdot (1.007 \pm 0.021) \times 10^{-5}}\right)} \\ &= 49.09 \pm 1.77\,\text{Hz} \end{aligned}$$

Theoretical damped frequency is

$$\begin{aligned} f_{theo} &= \left(\frac{1}{2 \cdot \pi}\right) \cdot \sqrt{\left(\frac{1}{L \cdot C}\right) - \left(\frac{\text{R}}{2 \cdot L}\right)^2} \\ &= \left(\frac{1}{2 \cdot 3.142 \times 10^0}\right) \cdot \sqrt{\left(\frac{1}{(1.044 \pm 0.072) \times 10^0 \cdot (1.007 \pm 0.021) \times 10^{-5}}\right) - \left(\frac{(1.794 \pm 0.013) \times 10^2}{2 \cdot (1.044 \pm 0.072) \times 10^0}\right)^2} \\ &= 47.14 \pm 1.59\,\text{Hz} \end{aligned}$$

Average current is calculated by

$$\begin{aligned} i_{av} &= \frac{V_R}{R_{res}} \\ &= \frac{-7.751 \times 10^{-1}}{-1.144 \times 10^0} \\ &= -1.598 \times 10^{-3}\,\text{A} \end{aligned}$$

Theoretical V_R is

$$\begin{aligned} V_R &= \left(\frac{(-V_0)}{L}\right) \cdot t \cdot \exp\left((-R) \cdot \frac{t}{2 \cdot L}\right) \cdot R_{res} \\ &= \left(\frac{\left(-3.860 \times 10^0\right)}{1.044 \times 10^0}\right) \cdot 6.000 \times 10^{-4} \cdot \exp\left(\left(-6.444 \times 10^2\right) \cdot \frac{6.000 \times 10^{-4}}{2 \cdot 1.044 \times 10^0}\right) \cdot 4.850 \times 10^2 \\ &= -8.940 \times 10^{-1}\text{V} \end{aligned}$$

Stored inductor energy is

$$E_L = 0.5 \cdot L \cdot (i_{av})^2 = 0.5 \cdot 1.044 \times 10^0 \cdot \left(-2.360 \times 10^{-3}\right)^2 \quad = 2.907 \times 10^{-6}\text{J}$$

Appendix B: Source code and raw data

The source code and raw data may be found it the public git repository:
`https://github.com/trolley-lorry/phys_bsc_data.git`.

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[1] *Physics 326 Electricity and Magnetism Laboratory Manual* (The University of Victoria, 2026).